Type MS · Volumes of revolution between two boundaries

Answer 1 of 10 · Question 2 part whole · 6 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

Number		
2	$4-x^2=x+2\Rightarrow x^2+x-2=0\Rightarrow (x-1)(x+2)=0\Rightarrow x=1,\ -2$	M1A1
	$V=\pi\int_{-2}^{1}\left(4-x^2\right)^2\,\mathrm{d}x-\pi\int_{-2}^{1}(x+2)^2\,\mathrm{d}x$	M1
	$V=\pi\int_{-2}^{1}x^4-9x^2-4x+12\,\mathrm{d}x=\pi\left[\frac{x^5}{5}-\frac{9x^3}{3}-2x^2+12x\right]_{-2}^{1}$	M1
	$V=\pi\left\{\left(\frac{1^5}{5}-3\times1^3-2\times1^2+12\times1\right)-\left(\frac{[-2]^5}{5}-3\times[-2]^3-2\times[-2]^2+12\times[-2]\right)\right\}$	M1
	$V=\frac{153\pi}{5}-9\pi$	
	$V=\frac{108\pi}{5}$	A1
	ALT	[M1
	$V=\pi\int_{-2}^{1}\left(4-x^2\right)^2\,\mathrm{d}x-\frac{\pi}{3}\times3^2\times3$	
	$V=\pi\int_{-2}^{1}\left(16-8x^2+x^4\right)\mathrm{d}x-[9\pi]=\pi\left[16x-\frac{8x^3}{3}+\frac{x^5}{5}\right]_{-2}^{1}-[9\pi]$	M1
	$V=\pi\left\{\left(16\times1-\frac{8(1)^3}{3}+\frac{(1)^5}{5}\right)-\left(16\times(-2)-\frac{8(-2)^3}{3}+\frac{(-2)^5}{5}\right)\right\}-[9\pi]$	M1
	$V=\frac{153\pi}{5}-9\pi$	
	$V=\frac{108\pi}{5}$	A1]
		Total 6 marks

Type MS · Volumes of revolution between two boundaries

Answer 1 of 10 · Question 2 part whole · 6 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Notes
M1	For equating the equation of the curve and the straight line, form a 3TQ with an acceptable attempt to solve the equation to find the x coordinates of the **two** points of intersection. [See General Guidance for the definition of an attempt] This mark is awarded for a complete method.
A1	For $x = 1$ and -2
Method 1	
M1	Uses the **correct** form for the volume of rotation. Ft their x coordinates used correctly. Everything must be correct for this mark including π
M1	For an attempt to integrate their expression for the volume of rotation for either the curve or the line (or even a combined expression). Do not accept a mixture of integration and differentiation. [See general guidance for minimum requirements for integration]. Award even if their expression is not squared. Ignore the absence of π or limits for this mark.
M1	For attempting to evaluate their integrated expression (which must be a changed expression) with their limits the correct way round. Substitution must be seen. Ignore the absence of π and ft their values of x for this mark
A1	For the correct exact value of $\frac{108\pi}{5}$ or exact equivalent.
Method 2	
M1	Uses the **correct** form for the volume of rotation of the curve minus the volume of the cone. The volume of the cone is: $\frac{1}{3}\times\pi\times('1'+2)^2\times('1'-'-2')$ Ft their x coordinates for the limits and for the dimensions of the cone. Everything must be correct for this mark including π
M1	For an attempt to integrate their expression for the curve. Do not accept a mixture of integration and differentiation. [See general guidance for minimum requirements for integration]. Award even if their expression is not squared, Ignore the absence of π or limits for this mark.
M1	For attempting to evaluate their integrated expression (which must be a changed expression) with their limits the correct way round. Substitution must be seen. Ignore the absence of π and ft their values of x for this mark
A1	For the correct exact value of $\frac{108\pi}{5}$ or exact equivalent.

Type MS · Volumes of revolution between two boundaries

Answer 1 of 10 · Question 2 part whole · 6 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Volumes of revolution between two boundaries

Answer 2 of 10 · Question 10 part b · 5 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

10 a	$(y-1)+y^2=11 \Rightarrow y^2+y-12=0$ or $x^2+\left(x^2+1\right)^2=11 \Rightarrow x^4+3x^2-10=0$	M1
	$(y-3)(y+4)=0$ or $x^2+5 \quad x^2-2 \quad =0$	M1
	$(y=3 \quad y=-4) \quad \Rightarrow x^2=2 \quad x^2=-5$	M1
	For stating the x coordinate of A is $-\sqrt{2}$ **and** the x coordinate of B is $\sqrt{2}$	A1 (4)
b	$\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(11-x^2)-(x^2+1)^2 \, dx$ or $\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(x^2+1)^2-(11-x^2)dx$	M1
	$\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(-x^4-3x^2+10) \, dx"$	A1ft
	$(\pi)\left[-\frac{x^5}{5}-x^3+10x\right]_{(-\sqrt{2})}^{(\sqrt{2})}$	M1
	$(\pi)\left[\left(-\frac{("\sqrt{2}")^5}{5}-("\sqrt{2}")^3+10"\sqrt{2}"\right)-\left(-\frac{("-\sqrt{2}")^5}{5}-("-\sqrt{2}")^3-10"\sqrt{2}"\right)\right]=63.98$	M1 A1 (5)
ALT		
	$\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(11-x^2) \, dx-\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(x^2+1)^2 \, dx$ or $\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(x^2+1)^2 \, dx-\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(11-x^2) \, dx$	{M1}
	$\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(11-x^2) \, dx-\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(x^4+2x^2+1) \, dx$	{A1ft}
	$(\pi)\left[11x-\frac{x^3}{3}\right]_{(-\sqrt{2})}^{(\sqrt{2})}-(\pi)\left[\frac{x^5}{5}+\frac{2x^3}{3}+x\right]_{(-\sqrt{2})}^{(\sqrt{2})}$	{M1}
	$(\pi)\left[\left(11("\sqrt{2}")-\frac{("\sqrt{2}")^3}{3}\right)-\left(11("-\sqrt{2}")-\frac{("-\sqrt{2}")^3}{3}\right)\right]$	
	$-(\pi)\left[\left(\frac{("\sqrt{2}")^5}{5}+\frac{2("\sqrt{2}")^3}{3}+"\sqrt{2}"\right)-\left(\frac{("-\sqrt{2}")^5}{5}+\frac{2("-\sqrt{2}")^3}{3}"-\sqrt{2}"\right)\right]$	{M1} {A1}
	$=63.98$	(5)
		Total 9 marks

Note: the candidate may also set up the integral as

$2\times\pi\int_0^{\sqrt{2}}(11-x^2)-(x^2+1)^2 \, dx$ or $2\times\pi\int_{-\sqrt{2}}^{0}(11-x^2)-(x^2+1)^2 \, dx$ or

$2\times\left(\pi\int_0^{\sqrt{2}}(11-x^2) \, dx-\pi\int_0^{\sqrt{2}}(x^2+1)^2 \, dx\right)$ *or* $2\times\left(\pi\int_{-\sqrt{2}}^{0}(11-x^2) \, dx-\pi\int_{-\sqrt{2}}^{0}(x^2+1)^2 \, dx\right)$

If it is clear that the integral(s) being used at any point needs to be multiplied by 2, all marks can be awarded, follow the same principles in the main or ALT schemes if the multiply by 2 is seen anywhere in the solution.

If multiply by 2 is nowhere present or implied by a final answer, the maximum mark is M0 A0 M1 M1 A0

Type MS · Volumes of revolution between two boundaries

Answer 2 of 10 · Question 10 part b · 5 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correctly rearranging to get $x^2 = y - 1$ and correctly substituting to get an equation of the form $ay^2 + by + c = 0 \quad a,b,c \neq 0$ **or** For correctly substituting $y = x^2 + 1$, an attempt to expand $(x^2+1)^2$ and attaining an equation of the form $dx^4 + ex^2 + f = 0 \quad d,e,f \neq 0$
	M1	For a complete attempt to solve either $ay^2 + by + c$ or $dx^4 + ex^2 + f$ by any valid method – see general guidance for definition of minimally acceptable attempt.
	M1	For correctly solving their quadratic equation in y or quartic equation in x **and** for obtaining a value(s) for x^2 or x. Award this mark if the candidate does not state or dismisses a solution where x^2 is negative.
	A1	For stating the x coordinate of A is $-\sqrt{2}$ **and** the x coordinate of B is $\sqrt{2}$ Decimal equivalent allowed.
(b)	M1	For $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) - (x^2+1)^2 \, dx$ or $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (x^2+1)^2 - (11 - x^2) dx$ Allow use of their limits from part a.
	A1ft	For expanding the bracket and simplifying to give $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (-x^4 - 3x^2 + 10) \, dx$ The ft is only for use of their limits from part a.
	M1	For correct integration of their expressions. There must be at least 3 terms overall and there must be a term in x^4 or y^4. It is not necessary for π or limits to be present for this mark.
	M1	For correct substitution of their limits seen into their changed expression. Each of the limits needs to be substituted into the expression correctly at least once. Allow a fully correct final answer to imply this mark. Brackets must be correct but can be recovered later.
	A1	For awrt 63.98 (Calc: 63.9775…) If a negative value is found and changed at the end to a positive value, this final A mark cannot be awarded.
ALT	M1	For $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) \, dx - \pi\int_{-\sqrt{2}}^{\sqrt{2}} (x^2+1)^2 \, dx$ or $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (x^2+1)^2 \, dx - \pi\int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) \, dx$ Allow use of their limits from part a.
	A1ft	For expanding the bracket and simplifying $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) \, dx - \pi\int_{-\sqrt{2}}^{\sqrt{2}} (x^4 + 2x^2 + 1) \, dx$ The ft is only for use of their limits from part a.
	M1	For correct integration of their expressions. There must be at least 3 terms overall and there must be a term in x^4 or y^4. It is not necessary for π or limits to be present for this mark.
	M1	For correct substitution of their limits seen into their changed expression. Each of the limits needs to be substituted into each expression correctly at least once. Allow a fully correct final answer, with correct integration shown, to imply this mark. Brackets must be correct but can be recovered later.
	A1	For awrt 63.98 (Calc: 63.9775…) If a negative value is found and changed at the end to a positive value, this final A mark cannot be awarded.

Type MS · Volumes of revolution between two boundaries

Answer 2 of 10 · Question 10 part b · 5 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Volumes of revolution between two boundaries

Answer 3 of 10 · Question 10 part d · 7 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 3 · find part d in this mark scheme extract

10 (a)	$f(1) = 32(1^3) - 33(1) + 1 = 0$ *	B1 cso (1)
(b)	$(x-1)(32x^2+32x-1)=0$ A correct method shown to solve a quadratic	M1
	e.g. $\frac{-32 \pm \sqrt{32^2+4\times32}}{64}$	M1
	$(x=1)$ or $\frac{-4+3\sqrt{2}}{8}$ or $\frac{-4-3\sqrt{2}}{8}$ Accept decimals correct to 3 sf e.g. = 0.0303… or = –1.03 …	A1 A1 (4)
(c)	$\sqrt{x} = \frac{1}{8x}$	M1
	$p = \frac{1}{4}$	A1 (2)

	Notes
(a)	
B1 cso	For substitution of 1 into $f(x)$ to obtain the given result
(b)	
M1	For $(x-1)(32x^2+32x-1)=0$
M1	A correct method shown to solve a quadratic. If an algebraic method is not shown then M0A0A0 is awarded.
A1	For $\frac{-4+3\sqrt{2}}{8}$ Allow 0.0303 or better
A1	For $\frac{-4-3\sqrt{2}}{8}$ Allow -1.03 or better
(c)	
M1	For equating the two equations
A1	For $p = \frac{1}{4}$

Type MS · Volumes of revolution between two boundaries

Answer 3 of 10 · Question 10 part d · 7 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

(d)	$\pi\int_{\frac{1}{4}}^{a} x\,\mathrm{d}x = \pi\left[\frac{x^2}{2}\right]_{\frac{1}{4}}^{a}$	M1
	$= \pi\left(\frac{a^2}{2} - \frac{1}{32}\right)$	A1
	$\pi\int_{\frac{1}{4}}^{a} \frac{1}{64x^2}\,\mathrm{d}x = \pi\left[-\frac{1}{64x}\right]_{\frac{1}{4}}^{a}$	M1
	$= \pi\left(\frac{1}{16} - \frac{1}{64a}\right)$	A1
	$= \pi\left(\frac{a^2}{2} - \frac{1}{32}\right) - \pi\left(\frac{1}{16} - \frac{1}{64a}\right) = \frac{27\pi}{64}$	dM1
	$32a^3 - 33a + 1 = 0$	A1
	So $a = 1$	A1 (7)
	Alternative Method	
	$\pi\int_{\frac{1}{4}}^{a}\left(x - \frac{1}{64x^2}\right)\mathrm{d}x = \frac{27\pi}{64}$	M1
	$\int_{\frac{1}{4}}^{a}\left(64x - \frac{1}{x^2}\right)\mathrm{d}x = 27$	A1
	$\left[32x^2 + \frac{1}{x}\right]_{\frac{1}{4}}^{a} = 27$	M1 A1
	$\left(32a^2 + \frac{1}{a}\right) - (2+4) = 27$	dM1
	$32a^3 - 33a + 1 = 0$	A1
	So $a = 1$	A1 (7) **[14]**

Type MS · Volumes of revolution between two boundaries

Answer 3 of 10 · Question 10 part d · 7 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

	Notes
(d)	
M1	For an attempt to integrate $\pi\int_{\frac{1}{4}}^{a} x\,\mathrm{d}x$ Ignore limits
A1	For $\pi\left(\frac{a^2}{2}-\frac{1}{32}\right)$
M1	For an attempt to integrate $\pi\int_{\frac{1}{4}}^{a}\frac{1}{64x^2}\,\mathrm{d}x$ Ignore limits
A1	For $=\pi\left(\frac{1}{16}-\frac{1}{64a}\right)$
M1	Dependant on at least one previous M mark being awarded - for subtraction of the two integrals.
A1	For $32a^3-33a+1=0$ oe
A1	For $a=1$
	Alternative
M1	For $\pi\int_{\frac{1}{4}}^{a}\left(x-\frac{1}{64x^2}\right)\mathrm{d}x=\frac{27\pi}{64}$ Ignore limits
A1	For $\int_{\frac{1}{4}}^{a}\left(64x-\frac{1}{x^2}\right)\mathrm{d}x=27$ Ignore limits
M1	For an attempt to integrate $\int_{\frac{1}{4}}^{a}\left(64x-\frac{1}{x^2}\right)\mathrm{d}x$ Ignore limits
A1	$\left[32x^2+\frac{1}{x}\right]_{\frac{1}{4}}^{a}=27$
M1	Dependant on at least one previous M mark being awarded - for correct substitution of the limits
A1	For $32a^3-33a+1=0$ oe
A1	For $a=1$

Type MS · Volumes of revolution between two boundaries

Answer 4 of 10 · Question 11 part b · 9 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

11 (a)	at P $\quad b=\sqrt{a-2} \Rightarrow b^2=a-2*$	M1A1cso [2]
(b)	At P $y=b \Rightarrow y^2=b^2 \Rightarrow y^2=a-2$	B1
	$V=\pi\int_a^{16}\left(\sqrt{x-2}\right)^2\,\mathrm{d}x-\pi\int_a^{16}(a-2)\,\mathrm{d}x=\pi\int_a^{16}(x-2)\,\mathrm{d}x-\pi\int_a^{16}(a-2)\,\mathrm{d}x$	M1
	$\Rightarrow \pi\int_a^{16}(x-a)\,\mathrm{d}x$ or $\pi\int_a^{16}\left(\sqrt{x-2}\right)^2\,\mathrm{d}x-\pi(a-2)(16-a)$	
	$50\pi=\pi\left[\frac{x^2}{2}-ax\right]_a^{16}$ or $50\pi=\pi\left[\frac{x^2}{2}-2x\right]_a^{16}-\pi(a-2)(16-a)$	depM1A1
	$50\pi=\pi\left[\left(\frac{256}{2}-16a\right)-\left(\frac{a^2}{2}-a^2\right)\right]$	depM1
	or $50\pi=\pi\left[\left(96-\frac{a^2}{2}+2a\right)-\left(18a-a^2-32\right)\right]$	
	$\Rightarrow a^2-32a+156=0$	M1
	$\Rightarrow (a-6)(a-26)=0 \Rightarrow a<16$ so $a=6$	
	$b^2=a-2 \Rightarrow b^2=4 \Rightarrow b=2$	M1A1 A1 [9]
		Total 11 marks

Type MS · Volumes of revolution between two boundaries

Answer 4 of 10 · Question 11 part b · 9 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Type MS · Volumes of revolution between two boundaries

Answer 4 of 10 · Question 11 part b · 9 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

Type MS · Volumes of revolution between two boundaries

Answer 4 of 10 · Question 11 part b · 9 marks

2020 JAN · Paper 01 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

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Type MS · Volumes of revolution between two boundaries

Answer 5 of 10 · Question 11 part whole · 8 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

Number		
11	$e^x = \frac{4}{e^x} \Rightarrow e^{2x} = 4$	M1
	$x = \frac{1}{2}\ln 4$ or $x = \ln 2$	A1
	$\pi\int_0^{\ln 2} e^{2x}dx + \pi\int_{\ln 2}^{a} 16e^{-2x}dx$	M1 M1
	$\pi\left[\frac{1}{2}e^{2x}\right]_0^{\ln 2} + \pi\left[-8e^{-2x}\right]_{\ln 2}^{a}$	M1
	$\pi\left(2-\frac{1}{2}\right)+\pi\left(-8e^{-2a}+2\right)=$	M1
	$\frac{7\pi}{2}-8\pi e^{-2a}$ $\quad a=2$ and $k=\frac{7\pi}{2}$	M1A1
		Total 8 marks

Type MS · Volumes of revolution between two boundaries

Answer 5 of 10 · Question 11 part whole · 8 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Guidance
M1	Sets $e^x = \frac{4}{e^x}$ and attempts to make x the subject A minimum of $e^{2x} = 4$ or $e^x = 2$ is required to be seen for this mark.
A1	For obtaining either $x = \frac{1}{2}\ln 4$ or $x = \ln 2$
M1	For stating $\pi\int_0^{\ln 2}\left(e^x\right)^2 dx = \pi\int_0^{\ln 2} e^{2x}dx$ using the limits of their ln 2 (which must be of the form ln k) and 0 correctly **Note:** Condone a missing π here if it is seen at the final M mark.
M1	For stating $\pi\int_{\ln 2}^{a}\left(4e^{-x}\right)^2 dx = \pi\int_{\ln 2}^{a} 16e^{-2x}dx$ using a and the limit of their ln 2 (which must be of the form ln k where k is consistent between the two integrals) correctly **Note:** Condone a missing π if it is seen at the final M mark.
M1	For an attempt to integrate **both** expressions obtaining: Either $\left[\frac{e^{2x}}{2}\right]$ or $\left[\frac{-16e^{-2x}}{2}\right]$ $\left(\text{condone } \left[\frac{16e^{-2x}}{2}\right]\right)$ For this mark ignore the absence of π or incorrect/absent limits [need not be simplified]
dM1	For substituting **their** limits correctly (where their ln 2 must be of the form ln k where k is consistent between the two integrals) into their integrated expression. For this mark ignore the absence of π $\pi\left(\frac{e^{2\ln 2}}{2} - \frac{e^0}{2}\right) + \pi\left(-16e^{-2a} - \frac{\left(-16e^{-2\ln 2}\right)}{2}\right) = \pi\left(2 - \frac{1}{2}\right) + \pi\left(-8e^{-2a} + 2\right)$ This mark is dependent on the previous M mark.
M1	For equating to the given expression for the volume and equating coefficients to find values for a and k. We must see π used here for the award of this mark. $\left(\pi\left(2 - \frac{1}{2}\right) + \pi\left(-8e^{-2a} + 2\right) = \frac{7\pi}{2} - 8\pi e^{-2a}\right)$ $\frac{7\pi}{2} - 8\pi e^{-2a} = k - 8\pi e^{-4} \Rightarrow k = \ldots,\ a = \ldots$
A1	For $a = 2$ and $k = \frac{7\pi}{2}$

Type MS · Volumes of revolution between two boundaries

Answer 5 of 10 · Question 11 part whole · 8 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

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Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 1 of 10 · find part b in this mark scheme extract

10(a)	At A:	
	$\left[y = 3(4-0)^{\frac{1}{2}} = 3\times 2 = \right] 6$	B1
	$\left[\frac{dy}{dx} = \right] 3\times\frac{1}{2}\times -1\times(4-x)^{-\frac{1}{2}} \left[= -\frac{3}{2}(4-x)^{-\frac{1}{2}}\right]$ oe	M1
	When $x = 0$:	
	$\left[\frac{dy}{dx} = -\frac{3}{2}(4-0)^{-\frac{1}{2}} = \right] -\frac{3}{4}$ oe	A1
	Equation of l:	
	eg $y-6 = -\frac{3}{4}(x-0)$ or $y = -\frac{3}{4}x+6$ or $4y = -3x+24$ oe	M1A1 [5]
(b)	When $y = 0$:	
	$\left[\text{Curve: } 0 = y = 3(4-x)^{\frac{1}{2}} \Rightarrow x = \right] 4$	B1ft
	$\left[\text{Line: } 4\times 0+3x-24 = 0 \Rightarrow x = \right] 8$	
	$\left[V = \right]\pi\int_0^{"8"}\left("-\frac{3}{4}x+6"\right)^2 dx - \pi\int_0^{"4"}\left(3\sqrt{4-x}\right)^2 dx$ or	M1
	$\left[V = \right]\frac{1}{3}\times\pi\times 6^2\times"8"-\pi\int_0^{"4"}\left(3\sqrt{4-x}\right)^2 dx$	
	$\left[= \pi\int_0^{"8"}\left(\frac{9}{16}x^2-9x+36\right)dx - \pi\int_0^{"4"}(36-9x)\,dx\right]$	
	$= \left[\pi\right]\left[\frac{3}{16}x^3-\frac{9}{2}x^2+36x\right]_{[0]}^{["8"]} - \left[\pi\right]\left[36x-\frac{9x^2}{2}\right]_{[0]}^{["4"]}$	M1M1 A1ft
	$\left[= \pi\left[\frac{3}{16}\times"8"^3-\frac{9}{2}\times"8"^2+36\times"8"\right]-\pi\left[36\times"4"-\frac{9\times"4"^2}{2}\right]\right]$	
	$\left[= 96\pi-72\pi = \right] 24\pi$	[5]
		Total 10 marks

Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 2 of 10 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct unsimplified vector in **a** and **b** for $\overrightarrow{BC}$ or for $\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$
	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BC}$
	M1	For $\overrightarrow{BC} = \frac{1}{2}\overrightarrow{AD}$ oe (note accept eg $2BC = AD$ ie without vector notation for this mark) Accept equivalences – see example MS.
	A1	For the correct conclusion, with all previous marks gained. Ignore work on other vectors. Do not accept eg $2BC = AD$ without vector notation for this mark Note B0 B1 M1 A0 is a potential score. $2BC = AD$ followed by eg $10\mathbf{a} + 6\mathbf{b} = 10\mathbf{a} + 6\mathbf{b}$ will get Aq
Watch for equivalences such as working out vector $\overrightarrow{CB}$ and stating $\overrightarrow{CB} = -\frac{1}{2}\overrightarrow{AD}$ Permit eg $\frac{\overrightarrow{AD}}{\overrightarrow{BC}} = 2$		
(b)	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BD}$
	B1ft	For applying Pythagoras theorem fully and correctly to find the length BD Ft any vector believed to be their $\overrightarrow{BD}$. $8^2 + 2^2$ alone is insufficient.
	M1	For using their vector for $\overrightarrow{BD}$ with their length BD to attempt to form a unit vector
	A1	For the correct unit vector.

Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 3 of 10 · find part b in this mark scheme extract

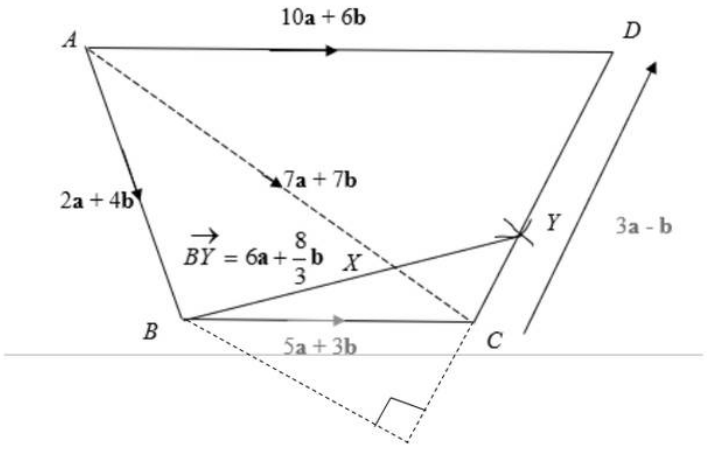
(c)(i)	$\left[\left[\overrightarrow{CY}=\right]\frac{1}{3}(3\mathbf{a}-\mathbf{b})=\mathbf{a}-\frac{1}{3}\mathbf{b}$ oe eg $\overrightarrow{YC}$ or $\left[\left[\overrightarrow{YD}=\right]\frac{2}{3}(3\mathbf{a}-\mathbf{b})=2\mathbf{a}-\frac{2}{3}\mathbf{b}$ oe eg $\overrightarrow{DY}$ Note: $\overrightarrow{BC}=5\mathbf{a}+4\mathbf{b}$ $\left[\overrightarrow{BY}=5\mathbf{a}+3\mathbf{b}+\mathbf{a}-\frac{\mathbf{b}}{3}=\right]6\mathbf{a}+\frac{8}{3}\mathbf{b}\Rightarrow$ $\left[\overrightarrow{BX}=\right]\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$ $\left[\overrightarrow{BX}=\right]-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $\left[\overrightarrow{BX}=\right](5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})$ $\left[\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $(5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})\right]$ $\left[6\lambda=7\mu-2$ and $\frac{8}{3}\lambda=7\mu-4$ or $\frac{8}{3}\lambda=-7\alpha+3$ and $6\lambda=-7\alpha+5\right]$ $\left[\lambda=\frac{3}{5};\mu=\frac{4}{5};\alpha=\frac{1}{5}\Rightarrow\overrightarrow{BX}=\right]\frac{18}{5}\mathbf{a}+\frac{8}{5}\mathbf{b}\Rightarrow[BX:XY=]3:2$	M1 dM1 ddM1 A1 [4]
ALT	First mark as main scheme. $\left[\overrightarrow{XY}=\right]\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$ $\left[\overrightarrow{XY}=\right]\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\left[\overrightarrow{XY}=\right]\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)$ $\left[\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)\right]$ $\left[6\beta=7\gamma+1$ and $\frac{8}{3}\beta=7\gamma-\frac{1}{3}$ or $\frac{8}{3}\beta=-7\delta+6+\frac{2}{3}$ and $6\beta=-7\delta+10-2\right]$ $\left[\beta=\frac{2}{5};\gamma=\frac{1}{5};\delta=\frac{4}{5}\Rightarrow\overrightarrow{XY}=\right]\frac{12}{5}\mathbf{a}+\frac{16}{15}\mathbf{b}[BX:XY=]3:2$	M1 dM1 ddM1 A1 [4]

Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 4 of 10 · find part b in this mark scheme extract

Part	Mark	Notes
(c)(i) Main and ALT	M1	Correct simplified vector in **a** and **b** for $\pm \overrightarrow{BY}$
	dM1	Correct vector in **a** and **b** with parameter using $\overrightarrow{CY}$ or $\overrightarrow{YC}$ for $\overrightarrow{BX}$ / $\overrightarrow{XB}$ or $\overrightarrow{XY}$ / $\overrightarrow{YX}$ **and** correct vector in **a** and **b** with a different parameter along a second distinct path using $\overrightarrow{AC}$ or $\overrightarrow{CA}$ for the same vector. Dependent on previous method mark.
	ddM1	For the correct simultaneous equations. Any one correct parameter or $\overrightarrow{BX} = \frac{18}{5}\mathbf{a} + \frac{8}{5}\mathbf{b}$ will imply this mark. Dependent on both previous method marks.
	A1	For the required ratio $BX : XY = 3 : 2$
Note, correct method can include the opposite signs in vectors multiplied by a parameter, as the parameter will just be the opposite sign when calculated and a fully correct solution is possible. Eg $\left[\overrightarrow{BX} = \right](5\mathbf{a}+3\mathbf{b}) + \alpha(7\mathbf{a}+7\mathbf{b})$ would just make $\alpha = -\frac{1}{5}$ Vectors questions notoriously have different methods. Correct answer, not from obviously incorrect working, full marks may be awarded **if method is shown.** (This is because the demand of this question says students must use a vector method). Min method must involve finding two correct different distinct relevant paths.		



Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 5 of 10 · find part b in this mark scheme extract

(ii)	$\left[\frac{CXY}{BXC}=\frac{2}{3}\text{ and }\frac{BAC}{BXC}=\frac{5}{1}\Rightarrow\right]\ \frac{CXY}{BAC}=\frac{2}{15}$ oe eg $\frac{BAC}{CXY}=\frac{15}{2}$	M1
Eg 1	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{CXY}{ABCD}=\frac{\frac{2}{15}}{3}\left[=\frac{2}{45}\right]$ oe eg $\frac{ABCD}{CXY}=\frac{3}{\frac{2}{15}}\left[=\frac{45}{2}\right]$	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
Eg 2	$\frac{CXY}{BCY}=\frac{2}{5}$ and $\left[\frac{BCY}{BCD}=\frac{1}{3}\right]\ \frac{BCY}{BCA}=\frac{1}{3}$ oe	M1
	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{BCY}{ABCD}=\frac{\frac{1}{3}}{3}\left[=\frac{1}{9}\right]$ oe and $\frac{CXY}{ABCD}=\frac{\frac{2}{5}}{9}\left[=\frac{2}{45}\right]$ oe	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
		Total 16 marks

Part	Mark	Notes
(ii)	M1	For any correct ratio, in any form, stated between a pair of triangles.
	M1	For any correct second ratio, in any form, stated between a different pair of triangles.
	M1	For any correct ratio, in any form between a triangle and the trapezium.
	A1	For the correct ratio only.
Any one of the ratios listed may be implied by a correct answer with no obviously incorrect working leading to this answer. Examples of other correct ratios: $\frac{BCD}{ABCD}=\frac{1}{3}$ $\frac{ABX}{CXY}=\frac{1}{6}$ **To be checked** Not carrying incorrect ratios through.		

Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 6 of 10 · find part b in this mark scheme extract

Question	Scheme	Marks
10(a)	At A:	
	$\left[y = 3(4-0)^{\frac{1}{2}} = 3\times 2 = \right] 6$	B1
	$\left[\frac{dy}{dx} = \right] 3\times\frac{1}{2}\times -1\times(4-x)^{-\frac{1}{2}} \left[= -\frac{3}{2}(4-x)^{-\frac{1}{2}}\right]$ oe	M1
	When $x = 0$:	
	$\left[\frac{dy}{dx} = -\frac{3}{2}(4-0)^{-\frac{1}{2}} = \right] -\frac{3}{4}$ oe	A1
	Equation of l:	
	eg $y-6 = -\frac{3}{4}(x-0)$ or $y = -\frac{3}{4}x+6$ or $4y = -3x+24$ oe	M1A1 [5]
(b)	When $y = 0$:	
	$\left[\text{Curve: } 0 = y = 3(4-x)^{\frac{1}{2}} \Rightarrow x = \right] 4$	B1ft
	$[\text{Line: } 4\times 0+3x-24 = 0 \Rightarrow x =] 8$	
	$\left[V = \right]\pi\int_0^{"8"}\left("-\frac{3}{4}x+6"\right)^2 dx - \pi\int_0^{"4"}\left(3\sqrt{4-x}\right)^2 dx$ or	M1
	$\left[V = \right]\frac{1}{3}\times\pi\times 6^2\times"8" - \pi\int_0^{"4"}\left(3\sqrt{4-x}\right)^2 dx$	
	$\left[= \pi\int_0^{"8"}\left(\frac{9}{16}x^2 - 9x + 36\right)dx - \pi\int_0^{"4"}(36-9x)\,dx\right]$	M1M1 A1ft
	$= [\pi]\left[\frac{3}{16}x^3 - \frac{9}{2}x^2 + 36x\right]_{[0]}^{["8"]} - [\pi]\left[36x - \frac{9x^2}{2}\right]_{[0]}^{["4"]}$	
	$\left[= \pi\left[\frac{3}{16}\times"8"^3 - \frac{9}{2}\times"8"^2 + 36\times"8"\right] - \pi\left[36\times"4" - \frac{9\times"4"^2}{2}\right]\right]$	
	$\left[= 96\pi - 72\pi = \right] 24\pi$	[5]
		Total 10 marks

Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

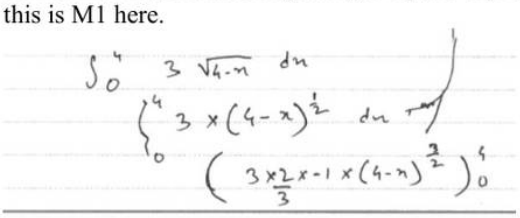
2025 NOV · Paper 02 · extract 7 of 10 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $[y =]\ 6$
	M1	For the correct derivative, doesn't need to be simplified.
	A1	For $-\frac{3}{4}$ oe If $-\frac{3}{4}$ oe from incorrect work – send to Review.
	M1	For correctly forming an equation of l using their values for y and $\frac{dy}{dx}$ Not a dependent mark, but there must be some attempt to have substituted $x = 0$ into a changed function to find a gradient. We will also accept if a candidate has done this and then found the negative reciprocal. If using $y = mx + c$, a correct rearrangement must be shown to find c and the equation then fully formed. If the rearrangement is not shown and the final equation is incorrect, c will need to be checked by examiners and must be correct for their values.
	A1	Award this mark for **any correct version** of the equation, simplified or not, rearranged or not, seen at any point. **It does not need to be given in the form stated by the question.**
In a complete break with convention, that must not be applied to any other question on this paper, examiners should: • mark crossed out work, even if replaced • mark multiple attempts (including crossed out work) • award the highest mark from all attempts For the usual conventions on marking crossed out work and multiple attempts, see General Guidance in the Mark Scheme.		

Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 8 of 10 · find part b in this mark scheme extract

(b)	**For part (b) – allow candidates to follow through $y = mx + c$ selected from (any multiple attempts) in part (a).**	
	B1ft	For $[x =]4$ **and** $[x =]$"8" ft the correct value of x for their $y = mx + c$ when $y = 0$ For example if $y = \frac{3}{4}x + 6$ is used $x = -8$ must be listed.
	M1	For a correct statement for the volume of revolution. Condone $[V =]\pi\int_0^4 \left(3\sqrt{4-x}\right)^2 - \pi\int_0^{"8"} \left("-\frac{3}{4}x+6"\right)^2 \, dx$ or $[V =]\pi\int_0^4 \left(3\sqrt{4-x}\right)^2 - \frac{1}{3}\times\pi\times 6^2\times"8"- \, dx$ This mark will ft their $y = mx + c$
	M1	For a minimally acceptable integration of their expressions, no power of x to decrease. There must be a minimum of 2 terms to integrate. π and limits do not need to be present. Simplification of terms after integration not necessary. If students have chosen to write down and find the volume of a cone, this is for a minimally acceptable integration of their $\left(3\sqrt{4-x}\right)^2$ Sometimes students are forgetting to square expressions. If you see this correct integration, this is M1 here. 
	M1	For a fully correct integration of $\left(3\sqrt{4-x}\right)^2$ π and limits do not need to be present. Simplification of terms not necessary.
	A1ft	For all of $\pi\int_0^{"8"} ("\text{their } mx+c")^2 \, dx - \pi\int_0^4 \left(3\sqrt{4-x}\right)^2 \, dx$ or condone $\pi\int_0^4 \left(3\sqrt{4-x}\right)^2 - \pi\int_0^{"8"} (\text{their } mx+c)^2 \, dx$ seen **and** fully and correctly expanded/integrated – terms do not have to be simplified. If writing down and using the formula for the volume of a cone, for their ft value of x from their equation, this must be evaluated (simplified) correctly as $r\pi \quad r \neq 0$ - examiners to check and there must be a fully correct integration of $\pi\int_0^4 \left(3\sqrt{4-x}\right)^2$. Examiners to check expansion/integration of $(\text{their } mx+c)^2$, π must be present at least once on each integration. We need to see the limits on all integration statements, ft their value of x for the intercept of the line but we do not need to see them substituted unless they are not present eg at the end of square brackets, as is convention. **We do not need to see and are not checking the final answer.**

Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 9 of 10 · find part b in this mark scheme extract

Note you may also see

$$\pi\int_0^4\left("-\frac{3}{4}x+6"\right)^2\,\mathrm{d}x-\pi\int_0^4\left(3\sqrt{4-x}\right)^2\,\mathrm{d}x\ +\pi\int_4^{"8"}\left("-\frac{3}{4}x+6"\right)^2$$

Type MS · Volumes of revolution between two boundaries

Answer 6 of 10 · Question 10 part b · 5 marks

2025 NOV · Paper 02 · extract 10 of 10 · find part b in this mark scheme extract

Question	Scheme	Marks
Look carefully at all diagrams, including the one provided. **Any contradictions with working, working takes precedence.**		

Type MS · Volumes of revolution between two boundaries

Answer 7 of 10 · Question 9 part whole · 10 marks

2024 NOV · Paper 01 · find part whole in this mark scheme extract

Number		
9	$\left(x=y^2+1 \qquad x=4-2y\right) \quad \left(y=\sqrt{x-1} \quad y=\dfrac{4-x}{2}\right)$	
	$4-2y=y^2+1 \qquad \text{or} \qquad \sqrt{x-1}=\dfrac{4-x}{2}$	M1
	$0=y^2+2y-3 \qquad 0=x^2-12x+20$	A1
	$0=(y+3)(y-1) \qquad 0=(x-2)(x-10)$	M1
	$(y=1,)\,x=2 \qquad x=2$	A1
	$\left(y^2=\left(2-\dfrac{x}{2}\right)^2=4-2x+\dfrac{x^2}{4}\right)$	
	$(0=x-1)\Rightarrow x=1 \qquad (2(0)+x-4=0)\Rightarrow x=4$	B1B1
	$\pi\int_{"1"}^{"2"}(x-1)\,dx+\pi\int_{"2"}^{"4"}\left(2-\dfrac{x}{2}\right)^2 dx \text{ or } \pi\int_{"1"}^{"2"}(x-1)\,dx+\dfrac{1}{3}\pi\times"1"^2\times("4"-"2")^2$	ddM1
	$\left[\pi\int_1^2(x-1)\,dx+\pi\int_2^4\left(4-2x+\dfrac{x^2}{4}\right)dx \text{ or } \dfrac{1}{3}\pi\times1^2\times2\right]$	
	$(\pi)\left[\dfrac{x^2}{2}-x\right]_{(1)}^{(2)}+(\pi)\left[\left(4x-\dfrac{2x^2}{2}+\dfrac{x^3}{12}\right)\right]_{(2)}^{(4)}$	M1
	$(\pi)\left[\left(\dfrac{"2"^2}{2}-"2"\right)-\left(\dfrac{"1"^2}{2}-"1"\right)\right]+(\pi)\left[\left(4("4")-\dfrac{2("4")^2}{2}+\dfrac{"4"^3}{12}\right)-\left(4("2")-\dfrac{2("2")^2}{2}+\dfrac{"2"^3}{12}\right)\right]$	M1
	$\left(\dfrac{1}{2}\pi+\dfrac{2}{3}\pi\right)$	
	$=\dfrac{7}{6}\pi \text{ oe}$	A1 [10]
		Total 10 marks

Type MS · Volumes of revolution between two boundaries

Answer 8 of 10 · Question 6 part b · 6 marks

2020 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

6 (a)	When $x = \frac{3}{2}$ $y = 3 - \frac{9}{4} = \frac{3}{4}$ and $2y - \frac{3}{2} = 0$	M1
	So $\left(\frac{3}{2}, \frac{3}{4}\right)$ lies on both the line and the curve	A1cso (2)
(b)	$(\pi)\int_0^{\frac{3}{2}}(2x - x^2)^2 dx$	M1
	$= (\pi)\int_0^{\frac{3}{2}}(4x^2 + x^4 - 4x^3)dx$	A1
	$= (\pi)\left[\frac{4x^3}{3} + \frac{x^5}{5} - x^4\right]_0^{\frac{3}{2}}$	dM1
	$= \frac{153(\pi)}{160}$	A1
	$\frac{153\pi}{160} - \frac{1}{3}\pi\left(\frac{3}{4}\right)^2\left(\frac{3}{2}\right) = \frac{27\pi}{40}$	ddM1A1cao (6) **[8]**
ALT:	$\pi\int_0^{\frac{3}{2}}\left((2x - x^2)^2 - \left(\frac{1}{2}x\right)^2\right)dx$	M1
	$\pi\int_0^{\frac{3}{2}}(\frac{15}{4}x^2 - 4x^3 + x^4)\,dx$	A1
	$\pi\left[\frac{5x^3}{4} - x^4 + \frac{x^5}{5}\right]_0^{\frac{3}{2}}$	dM1A1
	$\pi\left[\frac{135}{32} - \frac{81}{16} + \frac{243}{160}\right] = \frac{27\pi}{40}$	ddM1A1cao

(a)

M1 Attempt to show that $\left(\frac{3}{2}, \frac{3}{4}\right)$ lies on the curve and the line. Any valid method including solving the equations allowed.

A1cso Appropriate conclusion following correct work. Verification, as shown, needs a conclusion.

Solving the equations to obtain $\frac{x}{2} = 2x - x^2$ or $y = 4y - 4y^2$ and hence coordinates of A needs no conclusion. M1 for reaching coords, A1 for correct coords (decimals allowed)

Type MS · Volumes of revolution between two boundaries

Answer 8 of 10 · Question 6 part b · 6 marks

2020 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks
(b)	**Algebraic integration must be seen – otherwise no marks.** The first 4 marks can be awarded with or without π provided the work is consistent. The first 3 marks can be awarded if no limits are shown.	
M1	Correct integral, with or without π. Limits may be missing – ignore any shown.	
A1	Square the bracket correctly.	
dM1	Attempt the integration of their integrand. The power of at least one term should increase and no power should decrease. Ignore limits.	
A1	Substitute the correct limits and obtain $\frac{153}{160}$ or $\frac{153\pi}{160}$ (0.95625(pi))	
ddM1	Subtract the volume of the cone from their previous answer. Both terms to include π	
A1cao	Correct final answer (0.675pi)	
ALT:	**See above for general instructions re integration**	
M1	Integral must be the difference of 2 squared terms	
A1	Correct integrand after squaring, need not be simplified	
dM1	Attempt the integration of their integrand. The power of at least one term should increase and no power should decrease.	
A1	Correct result	
ddM1	Substitute their limits	
A1cao	Correct final answer.	

Type MS · Volumes of revolution between two boundaries

Answer 8 of 10 · Question 6 part b · 6 marks

2020 JAN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$ar^7 \quad 1152$	

Type MS · Volumes of revolution between two boundaries

Answer 9 of 10 · Question 5 part whole · 8 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

5	$\sqrt{2x+6}=\frac{x}{3}+1\Rightarrow 2x+6=\frac{x^2}{9}+\frac{2x}{3}+1\Rightarrow \frac{x^2}{9}-\frac{4x}{3}-5=0$	M1
	$\Rightarrow x^2-12x-45=0\Rightarrow (x+3)(x-15)=0\Rightarrow x=-3,15$	M1A1
	[When $x=15,\quad y=6$]	
	$(V=)\pi\int_{'-3'}^{'15'}(2x+6)\,dx-\frac{1}{3}\times\pi\times'6'^2\times('15'-'-3')$	M1M1
	$(V=)(\pi)\left[\frac{2x^2}{2}+6x\right]_{'-3'}^{'15'}-216\pi$	M1
	$(V=)\pi\left[\left(\frac{2\times'15'^2}{2}+6\times'15'\right)-\left(\frac{2\times('-3')^2}{2}+6\times('-3')'\right)\right]-216\pi(=324\pi-216\pi)=108\pi$	M1A1
ALT last 5 marks	OR	[8]
	$(V=)\pi\int_{'-3'}^{'15'}(2x+6)\,dx-\pi\int_{'-3'}^{'15'}\left(\frac{x^2}{9}+\frac{2x}{3}+1\right)dx$ or $\pi\int_{'-3'}^{'15'}(2x+6)\,dx-\pi\int_{'-3'}^{'15'}\left(\frac{x}{3}+1\right)^2dx$	M1M1
	$(V=)(\pi)\left[\frac{2x^2}{2}+6x\right]_{'-3'}^{'15'}-\pi\left[\frac{x^3}{9\times3}+\frac{2x^2}{3\times2}+x\right]_{'-3'}^{'15'}$	M1
	$V=\pi\left[\left(\frac{2\times'15'^2}{2}+6\times'15'\right)-\left(\frac{2\times('-3')^2}{2}+6\times('-3')\right)\right]$	
	$-\pi\left[\left(\frac{'15'^3}{9\times3}+\frac{2\times'15'^2}{3\times2}+15\right)-\left(\frac{('-3')^3}{9\times3}+\frac{2\times('-3')^2}{3\times2}-'3'\right)\right]=108\pi$	M1A1
		Total 8 marks

Type MS · Volumes of revolution between two boundaries

Answer 9 of 10 · Question 5 part whole · 8 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Notes
M1	Sets the equation of the curve = equation of line and attempts to form a 3TQ, which must include an attempt to expand $\left(\frac{1}{3}x+1\right)\left(\frac{1}{3}x+1\right)$. Allow only 1 error in processing overall. Accept as a minimum for this mark: $\frac{x^2}{9}\pm Jx-K=0$ or $x^2\pm Lx-M=0$ $J,K,L,M>0$
M1	Solves their 3TQ using a full and minimally acceptable method (see general guidance) to find two values. They must show their method if using an incorrect 3TQ. Note this is not a dependent mark, so the candidate can solve a 3TQ not of the form required for the first M mark.
A1	$x=-3, 15$ M1 M1 A1 may be awarded for both correct values given without working.
M1	For the correct expression for the volume of rotation under the curve including π and ft their $x=-3, 15$.
M1	For the correct expression for the volume of the cone. Ft their $x=-3, 15$ and their value of y only (must be clear they are using a value of y. '6' cannot be 15 or – 3). $\frac{1}{3}\times\pi\times'6'^2\times('15'-'-3')$
M1	For an attempt to integrate their expression for the curve. Minimally acceptable attempt (see general guidance) – no power of x to decrease. π and limits don't need to be present.
M1	For substitution of their limits into any changed expression representing their volume of the curve and evaluation of the final volume using their volume of the cone. Not a dependent mark but substitution **must** be into their changed expression. This mark may be awarded for the work being done in parts and subtracted in any order at any point, but **there must be a subtraction.** This M mark may be implied by a correct final answer.
A1	For $V=108\pi$ **This mark may not be awarded if the subtraction is the incorrect way around and the sign of the answer is simply changed at the end.**
ALT	last 5 marks
M1	For the correct expression for the volume of rotation under the curve including π and ft their $x=-3, 15$
M1	For the correct expression for the volume of rotation under the straight line including π and ft their $x=-3, 15$
Where candidates have combined the 2 expressions, M1 M1 can be awarded for $\pm\pi\int_{-3}^{15}\left(\frac{x^2}{9}-\frac{4x}{3}-5\right)\mathrm{d}x$ **or** for a clear attempt to combine the correct two expressions, even if the following simplification is incorrect. [Note for final A mark correct integral is $\pi\int_{-3}^{15}\left(-\frac{x^2}{9}+\frac{4x}{3}+5\right)\mathrm{d}x$	
M1	For an attempt to integrate the expression for **either** the curve **or** the line **or** what they intend to be **their** combined expression. Minimally acceptable attempt (see general guidance) – no power of x to decrease. π and limits do not need to be present for this mark.
M1	For substitution of their limits into any changed expressions representing their volume of the line and the curve and evaluation of the final volume or their combined expression. There must be at least one correct substitution of both limits into each expression. Not a dependent mark but substitution **must** be into changed expressions. This mark may be awarded for the work being done in parts and subtracted in any order at any point, but **there must be a subtraction.** This M mark may be implied by a correct final answer.
A1	For $V=108\pi$ **This mark may not be awarded if the subtraction is the incorrect way around and the sign of the answer is simply changed at the end.**

Type MS · Volumes of revolution between two boundaries

Answer 9 of 10 · Question 5 part whole · 8 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks
PLEASE REMEMBER TO CHECK DIAGRAM FOR RELEVANT LENGTHS AND WORKING **Throughout the question where lengths are in brackets eg (AC =), this means you don't have to see AC =.** **But the mark cannot be awarded for a clearly incorrect method eg $VO = \sqrt{x^2 + x^2}$ for the first method mark in (a) scores M0**		

Type MS · Volumes of revolution between two boundaries

Answer 10 of 10 · Question 10 part c · 5 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

10 (a)(i)	$\cos(A+A)=\cos A\cos A-\sin A\sin A\rightarrow\cos 2A=\cos^2 A-\sin^2 A$ $\cos 2A=1-\sin^2 A-\sin^2 A\Rightarrow\cos 2A=1-2\sin^2 A$ * $\cos 2A-1=-2\sin^2 A$ or $1-\cos 2A=2\sin^2 A$ $\sin^2 A=\frac{1-\cos 2A}{2}$ *	M1 M1 A1*cso [3]
/ALT (a)(i)	Starting from r.h.s of equation: $\cos(A+A)=\cos A\cos A-\sin A\sin A\rightarrow\cos 2A=\cos^2 A-\sin^2 A$ Gives $\frac{1-\cos 2A}{2}=\frac{1-(\cos^2 A-\sin^2 A)}{2}$ $=\frac{\sin^2 A+\sin^2 A}{2}=\sin^2$ as required.	M1 M1 A1*cso [3]
(ii)	$\left(\cos 2A=\cos^2 A-\sin^2 A\right.$ or $\left.\cos 2A=1-2\sin^2 A\right)$ $\cos 2A=1-\sin^2 A-\sin^2 A$ or $\cos 2A=1-2(1-\cos^2 A)$ $\rightarrow\cos 2A=2\cos^2 A-1$ $\cos 2A+1=2\cos^2 A$ $\cos^2 A=\frac{\cos 2A+1}{2}$ *	M1 A1*cso [2]
ALT (ii)	Starting from r.h.s of equation: $\cos 2A=\cos^2 A-\sin^2 A$ Gives $\frac{\cos 2A+1}{2}=\frac{\cos^2 A-\sin^2 A+1}{2}$ $=\frac{\cos^2 A+\cos^2 A}{2}=\cos^2 A$ as required.	 M1 A1*cso [2]
(b)	$\cos 2x=\sin x$ $1-2\sin^2 x=\sin x\rightarrow 0=2\sin^2 x+\sin x-1$ $0=(2\sin x-1)(\sin x+1)$ $\sin x=\frac{1}{2},(-1)\rightarrow x=$ $\left(\frac{\pi}{6},\frac{1}{2}\right)$	M1 M1 M1 A1 A1 [5]

Type MS · Volumes of revolution between two boundaries

Answer 10 of 10 · Question 10 part c · 5 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

(c)	$\pi\int_0^{\frac{\pi}{6}}\left(\cos^2 2x-\sin^2 x\right)dx$	B1ft
	$(\pi)\int_0^{\frac{\pi}{6}}\left(\frac{\cos 4x+1}{2}\right)-\left(\frac{1-\cos 2x}{2}\right)dx$	M1
	$\left[(\pi)\int_0^{\frac{\pi}{6}}\left(\frac{\cos 4x}{2}+\frac{\cos 2x}{2}\right)dx\right]$	
	$(\pi)\left[\frac{\sin 4x}{8}+\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{6}}$	M1
	$(\pi)\left(\frac{\sin 4\left(\frac{\pi}{6}\right)}{8}+\frac{\sin 2\left(\frac{\pi}{6}\right)}{4}\right)$	M1
	$\frac{3\sqrt{3}}{16}\pi$	A1 [5]
		Total 15 marks

Type MS · Volumes of revolution between two boundaries

Answer 10 of 10 · Question 10 part c · 5 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
10 (a)(i)	M1	Correct use of the addition formula as shown leading to $\cos 2A = \cos^2 A - \sin^2 A$
	M1	Correct substitution of $1-\sin^2 A$ to get $\cos 2A = 1 - 2\sin^2 A$
	A1*cso	Fully correct solution with the minimum steps shown, no errors.
ALT (a)(i)	M1	Correct use of the addition formula as shown leading to $\cos 2A = \cos^2 A - \sin^2 A$
	M1	Correct substitution into the right hand side of the equation.
	A1*cso	Rearrangement of resultant expression to reach left hand side of original equation., without errors..
(ii)	M1	Replacement of $\sin^2 A$ with $1-\cos^2 A$ in an expression from their part (i)
	A1*cso	Rearrangement of a correct equation to attain the given result, minimum steps shown, no errors.
ALT (ii)	M1	Replacement pf cos2A with expression from (i)
	A1*cso	Rearrangement of resultant expression to reach left hand side of original equation, without errors.
(b)	M1	Correctly equates the 2 expressions
	M1	Correctly uses the identity for cos2x and rearranges to an equation of the form $0 = a\sin^2 x + b\sin x + c \qquad a,b,c \neq 0$
	M1	Uses a full method to solve their equation of the form $0 = a\sin^2 x + b\sin x + c \qquad a,b,c \neq 0$ to find the inverse sine to arrive at a value for x. See general guidance for the solution of a 3TQ.
	A1	For the correct value of x
	A1	For correct coordinates, allow omission of brackets or x = and y = . **SC**: Allow A1 where candidate works in degrees and gets x = 30 degrees
(c)	B1ft	For the integral shown, ft their limits only.
	M1	An attempt to replace $\cos^2 2x$ **and** $\sin^2 x$ with an identity from part (a). At least one replacement must be fully correct.
	M1	At least one term correctly integrated.
	M1	Substitution of their limits clearly seen into a changed expression. This can be implied by a correct answer. Where the expression is not correct, or their limits are not correct, the substitution must be seen into at least half of their terms.
	A1	For the correct answer. Withold this mark if the candidate works in degrees rather than radians

Type MS · Volumes of revolution between two boundaries

Answer 10 of 10 · Question 10 part c · 5 marks

2025 JUN · Paper 01 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

Question number	Scheme	Marks